HW2 313551170 王書程

§ Prove Beta-Binomial conjugation

likelihood : $m \sim \text{Binomial}(N, p)$, $p(m|N,p) = \binom{N}{m} p^m (1-p)^{N-m}$

prior : $p \sim \text{Beta}(a,b)$, $p(p|a,b) = \frac{\gamma(a+b)}{\gamma(a)\gamma(b)} p^{a-1}(1-p)^{b-1}$

posterior : $p(p|N, m, a, b)$

$$= \frac{\text{likelihood} \times \text{prior}}{\text{marginal}} = \frac{p(m|N,p) \times p(p|a,b)}{p(D)}$$

$$= \frac{\binom{N}{m} p^m (1-p)^{N-m} \frac{\gamma(a+b)}{\gamma(a)\gamma(b)} p^{a-1} (1-p)^{b-1}}{\int_0^1 \binom{N}{m} \theta^m (1-\theta)^{N-m} \frac{\gamma(a+b)}{\gamma(a)\gamma(b)} \theta^{a-1} (1-\theta)^{b-1} d\theta}$$

$$= \frac{p^{m+a-1} (1-p)^{N-m+b-1}}{\int_0^1 \theta^{m+a-1} (1-\theta)^{N-m+b-1} d\theta}$$

By $\text{Beta}(a+m, N-m+b)$

$$\int_0^1 p(\theta|a+m, N-m+b)\, d\theta = 1$$

$$\Rightarrow \int_0^1 \frac{\gamma(a+b+N)}{\gamma(a+m)\gamma(N-m+b)} \theta^{a+m-1} (1-\theta)^{N-m+b-1} d\theta = 1$$

$$\Rightarrow \int_0^1 \theta^{a+m-1} (1-\theta)^{N-m+b-1} d\theta = \frac{\gamma(a+m)\gamma(N-m+b)}{\gamma(a+b+N)}$$

$$= \frac{\gamma(a+b+N)}{\gamma(m+a)\gamma(N-m+b)} p^{m+a-1} (1-p)^{N-m+b-1}$$

$\sim \text{Beta}(a+m, N-m+b)$